Integrated Science 1

OPTIMA ACADEMY ONLINE · FAMILY COURSE OVERVIEW · GRADES 9–10 · FULL YEAR

~4¼ hrs per week · 50 min/day, M–F

About this overview. This document is an example of how the course might be taught. More investigations and examples are approved for this course than appear here, so scholars may see some variation in labs, activities, and projects from course to course, and between the live school and On-Demand. Each teacher also leans into their own strengths and passions; the structure, expectations, and time commitment below are representative of the course as designed.

Integrated Science introduces scholars to foundational concepts in physical, life, and earth sciences through observation, experimentation, and logical reasoning, with an emphasis on clear thinking, precise language, and evidence-based conclusions. Aligned with Optima's classical approach and enriched by immersive VR and AI-supported explorations, the course builds genuine scientific literacy while cultivating wonder, order, and intellectual virtue in the study of the natural world. The year sweeps across the sciences: from atoms and the chemistry of life, through cells, genetics, and evolution, to ecology, geology, motion, energy, and the universe itself.

THE YEAR AT A GLANCE

Q1 Science, Matter, and the Chemistry of Life (9 weeks)
The nature of science and experimental design; atomic theory and the periodic table; macromolecules and the special properties of water; photosynthesis and cellular respiration.

Q2 Cells, Genetics, and Evolution (9 weeks)
Cell theory, structure, and transport; Mendel's laws, the cell cycle, mitosis and meiosis; the evidence for evolution, classification, and scientific explanations of the origin of life.

Q3 Ecology, Geology, and Motion (9 weeks)
Biogeochemical cycles and Earth's interacting systems; aquatic life, ecosystem change, and resources; Earth's layers and plate tectonics; Newton's first and second laws.

Q4 Energy, the Universe, and the Final (9 weeks)
Newton's third law; conservation of energy, heat, and waves; the Big Bang, the Sun, and the case for space exploration; a two-week review before the cumulative final exam.

A TYPICAL WEEK

- One content focus per week, at a steady pace through four nine-week quarters.
- Each week pairs instruction with a discussion focus: precise language and evidence-based reasoning about the week's concepts.
- Observation, experimentation, and data work throughout, with immersive explorations on Optima's VR campus.
- The year closes with a two-week review and a cumulative final exam.

TIME COMMITMENT

250 minutes / week

≈ 50 minutes a day, Monday–Friday

- Lessons, discussions, and investigation work fill most of the scheduled time.
- Review and exam weeks close each semester.

WHAT SCHOLARS NEED

- A computer with reliable internet and access to Canvas.
- A science notebook (digital or paper) for observations, data, and vocabulary.
- Access to Optima's VR campus for immersive explorations.

On-Demand (asynchronous). Scholars move through the course at their own pace, with due dates to keep the year on track. An Optima teacher is available throughout for support, feedback, and grading.

Live school. Live-school scholars take this same course on Optima's VR campus with a teacher and classmates, meeting daily as a core course.